

(21) Dokaži, da $\exists a, b$, da je vekt. polje $\vec{F} = \begin{bmatrix} 2x^a \sin z \\ 3y^b \sin z \\ (x^{a+1} + y^{b+1}) \cos z \end{bmatrix}$ potencialno.
Izračunaj potencial.

če polje pot, je $\text{rot } \vec{F} = \vec{0}$:

$$\text{rot } \vec{F} = \vec{\nabla} \times \vec{F} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x^a \sin z & 3y^b \sin z & (x^{a+1} + y^{b+1}) \cos z \end{vmatrix} =$$

$$= \begin{bmatrix} \frac{\partial}{\partial y} (x^{a+1} + y^{b+1}) \cos z - \frac{\partial}{\partial z} 3y^b \sin z \\ \frac{\partial}{\partial z} 2x^a \sin z - \frac{\partial}{\partial x} (x^{a+1} + y^{b+1}) \cos z \\ \frac{\partial}{\partial x} 3y^b \sin z - \frac{\partial}{\partial y} 2x^a \sin z \end{bmatrix} = \begin{bmatrix} (b+1)y^b \cos z - 3y^b \cos z \\ 2x^a \cos z - (a+1)x^a \cos z \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow b+1=3 \rightarrow b=2$$

$$a+1=2 \rightarrow a=1$$

$\rightarrow$ ker je $\mathbb{R}^3$ zvezdasto obl., je polje v primeru $a=1, b=2$ potencialno

$$\vec{F} = \begin{bmatrix} 2x \sin z \\ 3y^2 \sin z \\ (x^2 + y^3) \cos z \end{bmatrix} = \nabla u = \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \\ \frac{\partial u}{\partial z} \end{bmatrix}$$

$$\left. \begin{aligned} \frac{\partial u}{\partial x} &= 2x \sin z \rightarrow u = x^2 \sin z + C_1 \\ \frac{\partial u}{\partial y} &= 3y^2 \sin z \rightarrow u = y^3 \sin z + C_2 \\ \frac{\partial u}{\partial z} &= (x^2 + y^3) \cos z \rightarrow u = (x^2 + y^3) \sin z + C_3 \end{aligned} \right\} \underline{u = (x^2 + y^3) \sin z (+C)}$$

Cirkulacija je int. vekt. polja po sklenjeni krivulji. Vrednost je neodv. od izbire zač. točke, do predznaka nat. odv. od orientacije.

Greenova formula: D omejena ravn. obm. z odsekoma gladkim robom ∂D , ki je +orientiran

Za vekt. polje $\vec{R} = \begin{bmatrix} x \\ y \end{bmatrix}$, ki je zvezno na $\bar{D} = D \cup \partial D$, in zvezno odv. na D je

$$\oint_{\partial D} \vec{R} \cdot d\vec{r} = \iint_D \left(\frac{\partial y}{\partial x} - \frac{\partial x}{\partial y} \right) dx dy$$

(22) Izračunaj $\oint_K \vec{R} \cdot d\vec{r}$, $\vec{R} = \begin{bmatrix} -y/2 \\ x/2 \end{bmatrix}$, K poz. orientiran rob prav. 6-kotnika s središčem

v $(0,0)$, in enim ogliščem v $(1,0)$.

$$\frac{\partial y}{\partial x} - \frac{\partial x}{\partial y} = 1$$

$$\oint_K \vec{R} \cdot d\vec{r} = \iint_D dx dy = 6 \cdot \frac{\sqrt{3}}{22} = \underline{\underline{\frac{3\sqrt{3}}{2}}}$$

Gaussov izrek: D omejeno prost. obm. z odsekoma gladkim robom, normala

kašče navzven. Tedaj t. zv. odv. vekt. polje $\vec{R}$ velja:

$$\iint_{\partial D} \vec{R} \cdot d\vec{S} = \iint_{\partial D} \langle \vec{R}, \vec{N} \rangle dS = \iiint_D \operatorname{div} \vec{R} dV$$

(23) Izračunaj pretok vekt. polja $\vec{R} = \begin{bmatrix} x(y^2+z^2) \\ y(x^2+z^2) \\ z(x^2+y^2) \end{bmatrix}$ skozi rob ehotske krogle, orientiran navzven.

$$\operatorname{div} = \vec{\nabla} \cdot \vec{R} = (y^2+z^2) + (x^2+z^2) + (x^2+y^2) = 2x^2 + 2y^2 + 2z^2$$

$$\iint_{\partial D} \vec{R} \cdot d\vec{S} = 2 \iiint_D (x^2+y^2+z^2) dV = 2 \int_0^\pi \int_0^{2\pi} \int_0^1 r^4 \sin \theta dr d\varphi d\theta = 2 \cdot 2\pi \cdot \cos \theta \Big|_0^\pi \cdot \frac{1}{5} r^5 \Big|_0^1 = \frac{8}{5} \pi$$

$x^2+y^2+z^2 \leq 1$

Skladna orient.: S omejena odsekoma gladka pl. v prostoru z odsekoma glatkim robom ∂S , S in ∂S skladno orientirani, če imamo, ko se sprehajamo po robu v smeri orientacij in obrnjeni tako kot $\vec{N}$ plosker na levi.

Stokesov izrek: S , ∂S skladno orientirana, $\forall$ zu. odv. vekt. polje velja

$$\oint_{\partial S} \vec{R} \cdot d\vec{r} = \iint_S \langle \text{rot} \vec{R}, \vec{N} \rangle dS$$

(2.1) Izr. pretek vekt. polja $\vec{R} = \begin{bmatrix} e^{x^2+xy} \\ z + \sin^5(x^5) \\ e^{z^2+2x+y+\cos z} \end{bmatrix}$ po krivulji $\begin{matrix} x^2+y^2=1 \\ z=xy \end{matrix}$,

orientirano tako, da se, če jo gledamo od zgoraj, vrti v + smeri.

$$\vec{\nabla} \times \vec{R} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{x^2+xy} & z + \sin^5(x^5) & e^{z^2+2x+y+\cos z} \end{vmatrix} =$$

$$= \begin{bmatrix} 0 \\ -2 \\ x \end{bmatrix}$$

$\rightarrow$ za ploskev vzamemo $x^2+y^2 \leq 1$, $z=xy$ graf funke.

$$\rightarrow \text{normala } \vec{N} = \begin{bmatrix} f_x \\ f_y \\ 1 \end{bmatrix} = \begin{bmatrix} -y \\ -x \\ 1 \end{bmatrix}$$

$$\iint_{x^2+y^2 \leq 1} \langle \text{rot} \vec{R}, \vec{N} \rangle dS = \iint_{-1-\sqrt{1-y^2}}^{1-\sqrt{1-y^2}} x dx dy = \int_{-1}^1 (1-y^2) dy = 0$$

NDE

② Poiščite rešitev $y^2 = xy y' + g$; $y(1) = -2$. Katero je njeno max. def. obr.?

$$y' = \frac{y^2 - g}{xy} = \frac{1}{x} \left(y - \frac{g}{y} \right) \quad \frac{dy}{dx} = f(x)g(y)$$

$$\rightarrow f(x)dx = \frac{dy}{g(y)} \quad / \int$$

$$\ln|x| = \int \frac{y}{y^2 - 9} dy = \frac{1}{2} \ln|y^2 - 9| + C$$

$$y(1) = -2$$

$$\ln 1 = \frac{1}{2} \ln 5 + C$$

$$C = -\ln 5^{\frac{1}{2}}$$

$$y^2 - 9 = u$$

$$2y dy = du$$

$$\rightarrow \ln|y^2 - 9| = 2 \ln|x| + \ln 5 = \ln|5x^2|$$

$$|y^2 - 9| = 5x^2$$

$$\text{v okolici zač. točke: } y^2 - 9 < 0$$

$$\Rightarrow 9 - y^2 = 5x^2$$

$$y = -\sqrt{9 - 5x^2}$$

↑ iz zač. pog.

$$\text{def. zn} \quad |x| < \frac{3}{\sqrt{5}}$$

⑥ part. res. $xy' - y = x \operatorname{tg} \frac{y}{x}$, $y(3) = 8\pi$, max. def. obr.

$$z = \frac{y}{x} \rightarrow y = xz \rightarrow y' = z + xz'$$

$$z + xz' - z = \operatorname{tg} z$$

$$z' = \frac{dz}{dx} = \frac{\operatorname{tg} z}{x} \rightarrow \frac{dz}{\operatorname{tg} z} = \frac{dx}{x} \quad / \int$$

$$\ln|x| = \int \frac{\cos z dz}{\sin z} = \ln \frac{\sin z}{C'} \rightarrow \sin z = C'|x| = Cx$$

$$\sin z = u$$

$$\cos z dz = du$$

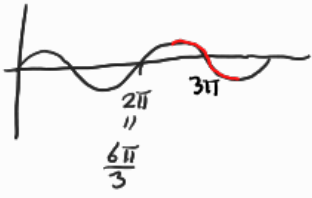
$$\sin \frac{8\pi}{3} = 3C$$

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} = 3C$$

$$C = \frac{1}{2\sqrt{3}}$$

$$\sin \frac{y}{x} = \frac{x}{2\sqrt{3}}$$

→ ko obrnemo sin, želimo dobiti del, ki vsebuje tačko $\frac{8\pi}{3}$



$$\rightarrow \frac{y}{x} = 3\pi - \arcsin \frac{x}{2\sqrt{3}}$$

$$y = \underline{\underline{3\pi x - x \arcsin \frac{x}{2\sqrt{3}}}}$$